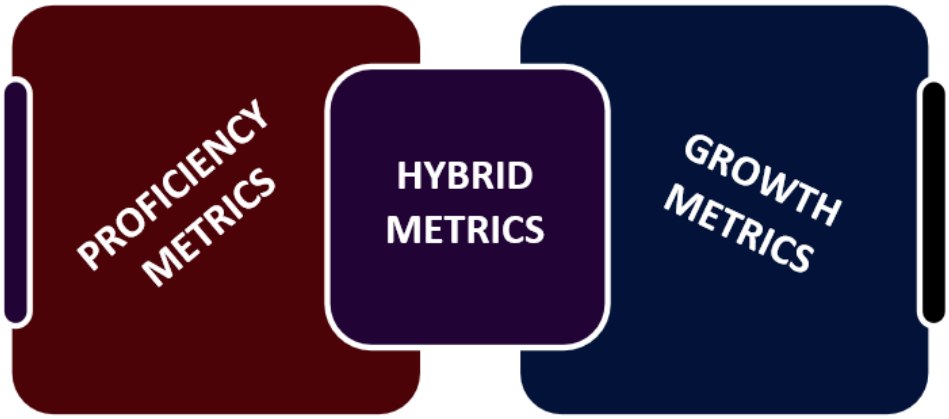
Section 6: Psychometrics, Machine Learning, & LMS

Oftentimes, psychometricians face the difficult task of identifying quantifiable input/output measures to develop fair, consistent, and high-quality assessments in education. Unlike employment assessments, where aptitude can be measured according to professional certifications, job analysis, NLP, and training matrices, educational assessments are a strong function of curriculum, lesson plan quality, test creation, conditioning, scaffolding, student progress, and student proficiency. Such a function requires the collation between data analytics (quantifiable measures) and observations/surveys (qualitative measures) to meet the needs of students. Although the qualitative measures are handled well through diagnostic assessments, formative assessments, and summative assessments, the quantitative measures have continuously posed problems due to the absence of standardized evaluation theories for proficiency, growth, and the hybrid of the two. (See the figure below)



EvalzNow will be the first and only software to provide proficiency metrics, hybrid metrics, and growth metrics based on cognitive science, mathematical modeling, and closed-loop brain interactions.

Capturing these metrics will lead to:

- The enhancement of item response theory (IRT) paradigms for analyzing test questions through the interaction of examinees. Proficiency metrics, growth metrics, and hybrid metrics can be incorporated into machine learning models, which utilize logistic regression.
- More effective descriptions of how quantifiers contribute to learning without weighted proficiency
- Improved algorithms for adaptive modules, as well as attained score captures in learning platforms and computerized testing
- Improved algorithms for AI assessment systems, especially for feedback and growth
- A comprehensive aptitude panel for students and curriculum design.

The absence of legitimate mathematical models continues to plague educational institutions that adopt AI machine learning (big data approaches), deep learning, and learning management systems (LMS).

Due to the absence of quantitative systems in education and industry, existing evaluation modules normally track random data sets (most not applicable), engagement levels (aimed at the highest % rather than the effective %), and lumped data sets (proficiency only). Lumped data sets, to begin, will result in one (1) proficiency metric that will not be sufficient for assessing student learning because student learning is a function of multiple proficiency measures and student growth.

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Any educator, who is considering machine learning or neural network software, should wonder:

- How does the software calculate growth targets for a given student?
- Can the software calculate growth targets without making a comparison to other students?
- Can the software calculate the % growth of an individual student from the beginning of the course to the end of the course?
- How many growth metrics and proficiency metrics does the software track?
- What proven theory is the assessment model derived from?
- What do the inputs and the outputs of the software mean practically?
- How reliable are the results? How do data analytics compare with qualitative assessments?
- Do you incorrectly use a diagnostic test, which is a proficiency test, to assess student growth? Why?
- How can the Adaptive Learning Module of the software work properly if there is not a sufficient number of proficiency metrics and growth metrics programmed into the Evaluation Module?
- What assumptions are programmed into the Evaluation Module and the Adaptive Learning Module?
- Does the Evaluation Module of the software treat STEM and non-STEM courses the same?
- Does the software incorrectly integrate a percentile model, knowing that no students are the same?
- What is the effectivity of AI training iterations targeting student performance being monitored/assessed?
- Does the neural network algorithm have parity issues for complex learning paradigms/techniques?
- Can the neural network find/create learning algorithms on its own – inclusive of external eval criteria?

These are critical questions that will help educators determine if the machine learning and/or neural network algorithm has circumvented the process of deriving a proficiency and growth model on fundamental theories. Using engagement levels as a means for student assessment, in continuance, leads to problems in AI learning as well. Data analysts, who normally generate educational algorithms without having in-depth knowledge of the student learning process, cognitive science, and educational pedagogy, do not realize that engagement levels are inversely proportional to challenge levels.

If the engagement levels are too high, then the challenge levels are too low; however, data analysts try to convince educators that high engagement is attributed to growth when that is not true.

If the engagement levels are too low, then the challenge levels are too high. Data analysts try to convince educators that low engagement means that the course is boring, which is not always the case.

Educators should encourage data analysts to find the maximum engagement level that results in the highest reasonable challenge level after looking into proficiency and growth metrics because engagement does not take precedence over proficiency and growth, and the level of engagement is a tertiary factor with low reliability. For instance, engagement levels can be highly skewed if a student takes multiple breaks, falls asleep, has distractions, faces easy questions that require a lot of work, takes strategic guesses, and so forth. Moreover, engagement levels do not determine whether a student goes to the next grade level.

This leads to the final thought on machine learning associated with optimization. If a machine-learning algorithm does not have a legitimate proficiency and growth model, then the Optimization Module will be rendered useless because:

- There will be no objective way to prove that the curriculum needs to be improved.
- There will be no objective way to prove that the curriculum is not biased toward specific student groups – normally, left-brained learners.
- There will be no objective way to prove that the curriculum did not cause student failure.

An evaluation software should never have a rift between theory and practice, and changing data sets renders existing logic obsolete. Put in another way, any software, which utilizes an illegitimate and/or an unreliable Evaluation Module and/or Optimization Module, will have to discard all existing data upon update. The existing data will continue to result in poor predictions, poor assumptions, poor student adaptations, poor data outputs, and so forth. This might cripple some software, especially since the maturation period for trial & error is long (> 2-5 years in some cases).

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EvalzNow addresses all of these potential shortcomings with a full-scale evaluation of student learning beginning on day 1. For deep learning assessments, the user should use the Theory & Practice Model or the Third Parameter Model.

Learning Management Systems (LMS), which are normally coupled with Content Management Systems (CMS), not only provide learning-centered instruction to employees/students but also foster global knowledge inclusion at a low expense to companies that must meet L&D requirements, regulatory requirements, continued education, workplace safety, philanthropy, etc. The strength of these systems in education and industry relies on the understanding that all training/instruction does not require traditional face-to-face settings. This understanding leads to three (3) adaptive choices of a given institution: traditional in-person delivery, blended delivery, and distance/virtual delivery. Regardless of the adoptive choice, EvalzNow can supplement and can even refine/improve the learning approach by:

1. Having the LMS provider to set curricula, to chart certification/career paths, to assign tutors based on hybrid metrics, proficiency metrics, and growth metrics comprehensively. LMS can only meet rigid/simple proficiency requirements.
2. Having the LMS provider to author/maintain/store content that encompasses all inputs and all outputs for a given mathematical model (I.E., CI/CO, T&P, or TPRM). LMS must have applicable and authentic content; however, the content does not always have the necessary components integrated into the framework for assessment. For instance, a language course (CI/CO) with a missing writing component or a Lean 6 (T&P) training segment with a missing practical component will lead to faulty assessments.
3. Having the LMS provider to consider Competency Accounting when administering examinations, self-checks, tests, workbook assignments, and quizzes for attainment. If done properly, all of the attained scores can be extracted from the LMS into EvalzNow for analysis.

Coupling robust stability criteria with performance data, EvalzNow can also determine the stability of interactions against proficiency (STA-P) and the stability of interactions against growth (STA-G) per performance point for those seeking deep evaluations of either students and/or workers.

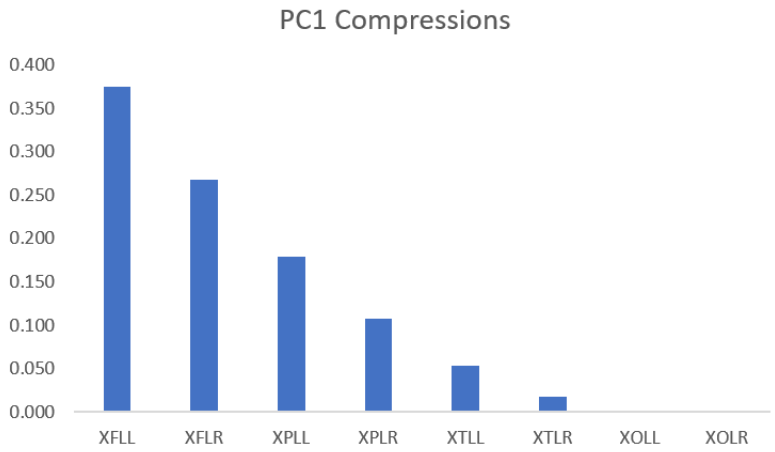
A neuroscience module has also been integrated to map performance point data to the major brain lobes and cognitive functions.

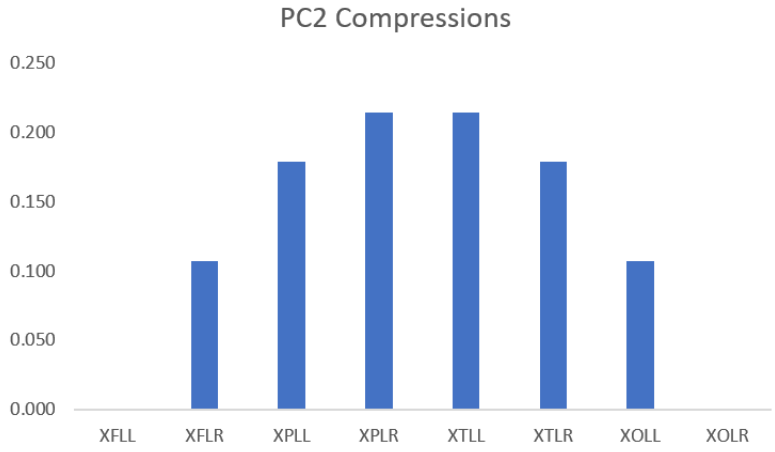
If the User wants to perform neuroscience studies as a function of competency, then it is best to use applicable unit types under Competency Accounting.

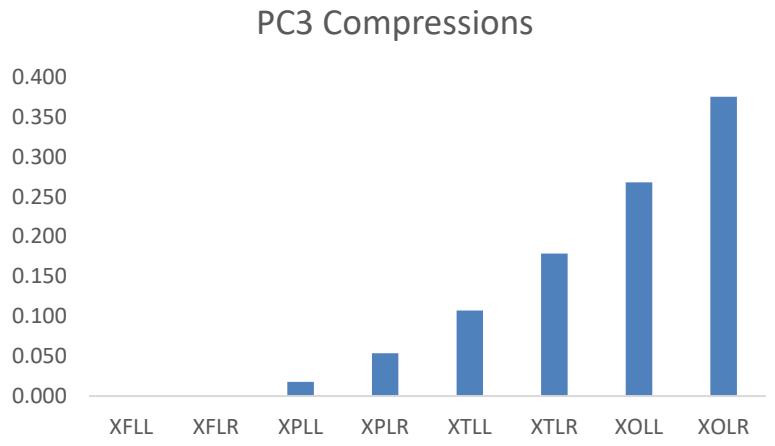
If the User wants to perform direct neuroscience studies – especially for neurotypical, neurodivergent, and neurodiverse cases – then it is best to apply the applicable capsule types under Capacity Accounting. Each major brain lobe is tracked separately, and then compressed into three (3) components using a centroid Principal Component Analysis (PCA) approach of all possible variable combinations to capture the maximum amount of variance (information).

If the User selects the Capacity – Isolated Capsule Type, then the Principal Component compression distributions will be as follows:

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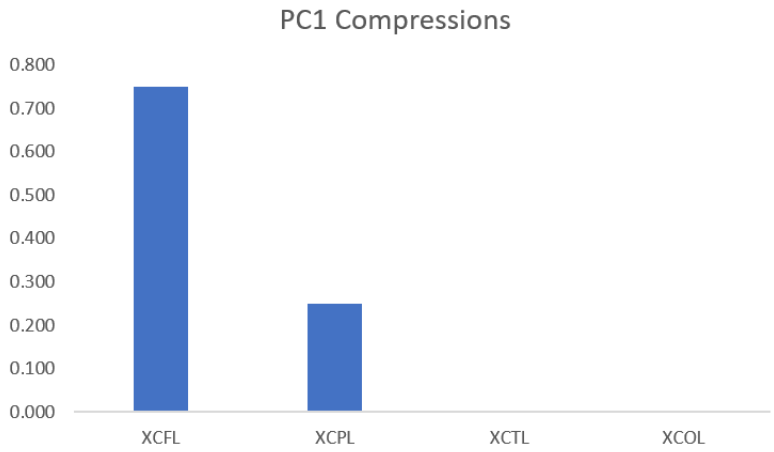


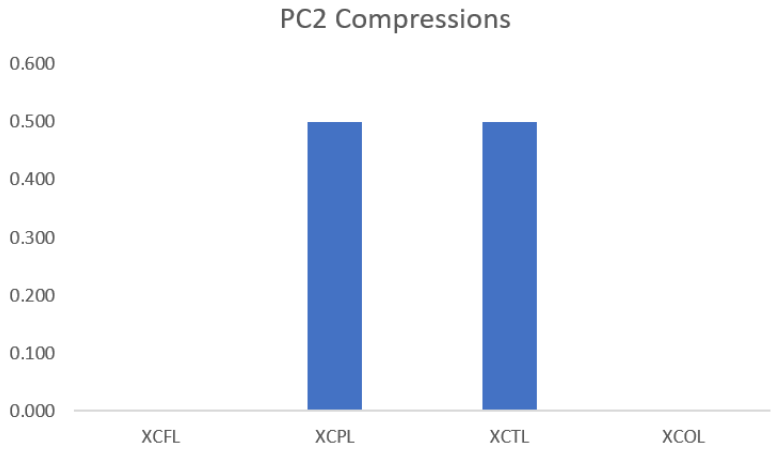


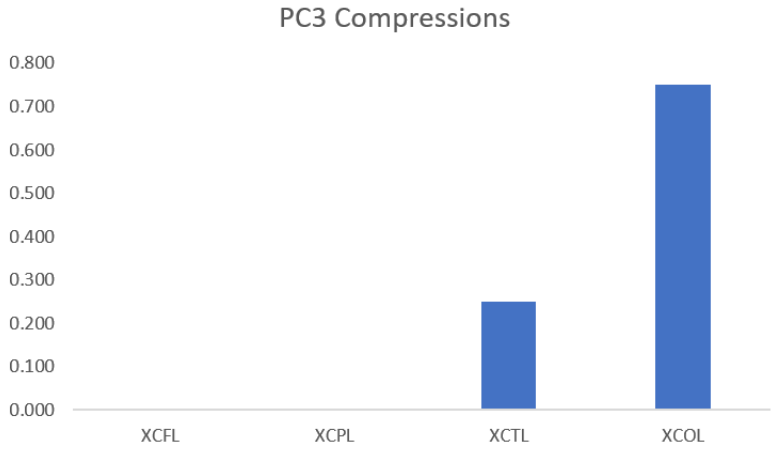


If the User selects the Capacity – Combined Capsule Type, then the Principal Component compression distributions will be as follows:

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Please note that outcomes from neurological brain scans can be integrated into the data pool; however, all assessments are limited to activity in the major lobes – frontal, parietal, temporal, and occipital.

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